

INTRODUCTION

Plants continuously interact with their environment, and one of the most important structures involved in this interaction is the stomata. Stomata are microscopic pores present on the leaf epidermis that regulate gaseous exchange and transpiration. According to Willmer and Fricker (1996), stomata play a central role in controlling water loss and carbon dioxide uptake, thereby directly influencing photosynthesis and plant productivity.

The study of stomatal frequency began with the pioneering work of Salisbury (1927), who demonstrated that stomatal density varies significantly among plant species and is closely related to ecological conditions. Later, Fahn (1990) explained that stomatal characteristics are important anatomical features useful in plant identification and classification.

The terms stomatal density (number of stomata per unit area) and stomatal index (percentage of stomata relative to total epidermal cells) are widely used parameters in plant anatomical studies. Meidner and Mansfield (1968) reported that stomatal index remains relatively constant for a species and can serve as a reliable taxonomic character. However, several researchers including Hetherington and Woodward (2003) have shown that environmental factors such as light, temperature, humidity, and carbon dioxide concentration can significantly influence stomatal development.

Studies by Franks et al. (2009) revealed that plants growing under stressful environmental conditions often show variations in stomatal size and density as an adaptive mechanism. Similarly, Pandey and Nagar (2002) reported variations in stomatal characteristics among different medicinal plants due to ecological conditions.

In addition to their physiological significance, stomatal characteristics are increasingly being used in applied fields such as ecology, taxonomy, palaeobotany, and environmental monitoring. Standard plant physiology texts such as Taiz and Zeiger (2015) have also emphasized the role of stomata in plant adaptation to environmental stress conditions including drought and high temperature.

AIM AND OBJECTIVES

Aim :-

The main aim of the present study is to conduct a comparative evaluation of stomatal density and stomatal index in selected plant species of the Akot region, Maharashtra, in order to understand their anatomical variation and ecological significance.

Objective :-

- 1.To select and identify 15–20 commonly available plant species from the Akot region.
- 2.To prepare temporary epidermal peels of leaves for microscopic observation.
- 3.To observe and record stomatal characteristics using a compound microscope.
- 4.To calculate the stomatal density (number of stomata per unit area).
- 5.To calculate the stomatal index using the standard formula.
- 6.To compare stomatal density and stomatal index among different plant species.
- 7.To analyze the relationship between stomatal characteristics and environmental conditions of the study area.
- 8.To generate baseline data that may be useful for future taxonomic and ecological studies.

REVIEW OF LITERATURE

Salisbury (1927)

Salisbury was one of the earliest scientists to study stomatal frequency (density) in plants. He examined a large number of plant species growing in different habitats such as dry, moist, and shaded areas. His research showed that plants growing in dry environments generally have fewer stomata, while plants in moist environments have more stomata.

Meidner and Mansfield (1968)

Meidner and Mansfield worked on the physiology of stomata. They studied how stomata open and close and how this process is controlled by factors such as light, water, CO₂ concentration, and hormones. Their book “Physiology of Stomata” explains the functional role of stomata in transpiration and gas exchange.

Willmer (1983)

Willmer studied stomatal index and explained that stomatal index is more reliable than stomatal density because it is less affected by environmental changes and more controlled by genetics. She suggested using stomatal index in taxonomy and ecological comparisons.

Woodward (1987)

Woodward studied the relationship between stomatal frequency and atmospheric CO₂ levels. He examined fossil leaves and modern plants and found that when CO₂ levels increase, stomatal density usually decreases.

Hetherington and Woodward (2003)

These scientists studied the genetic control of stomatal development. They explained how genes regulate the formation of stomata during leaf development.

Taiz and Zeiger (2010)

Taiz and Zeiger, in their famous book Plant Physiology, explained that stomatal density, stomatal index, and stomatal movement are closely related to plant water use efficiency and photosynthesis.

Parkhurst (1978)

Parkhurst studied the distribution of stomata on leaf surfaces (upper and lower epidermis) in many plant species. He reported that most dicot plants have more stomata on the lower surface, which reduces water loss.

Cutter (1978)

Cutter explained in his work on plant anatomy that stomatal number, size, and distribution vary with habitat. He compared xerophytes (dry plants) and mesophytes (moderate climate plants).